

From detector signals to $\Delta(x,Q^2)$: the reconstruction chain of the inclusive and coherent $\cos 2\varphi$ measurements

2026-08-24 · revised 2026-08-25 · companion to the $\cos 2\varphi$ projection report (money plots 5–7) and to plans/07 WP3 · computed with `polligen.reco`, `polligen.hfs` and `polligen.recopseudo` (`evgen/scripts/: reco_chain_figures.py`, `hfs_resolution.py`, `money_cos2phi_reco.py`, `money_cos2phi_coherent_reco.py`) · revision history in Appendix A

The projection report presents the double-helicity-flip structure function $\Delta(x,Q^2)$ as φ' pseudo-data and as extracted $x\Delta$ points. This note asks the experimentalist's question behind those plots: what does the detector actually record, event by event, in the inclusive and in the coherent intact-⁶Li measurement; how is the azimuth that carries the observable built from the reconstructed four-vectors and the known spin direction; which kinematic reconstruction assigns (x, Q^2) to an event; which estimator turns spin-sorted φ' histograms into an amplitude; and how the amplitude becomes Δ . It then states which of those operations the simulation performs (§5) and closes the chain at the reconstructed level (§7). Three of its results change the analysis design.

Three findings that change the analysis. (1) Three of the four inclusive sweet-spot bins of money plot 5 sit at $y = 0.010\text{--}0.025$, where the electron method gives $\delta y/y = 50\text{--}120\%$ even with a 1.2% calorimeter: x must come from the hadronic final state. At the documented 25% hadronic- y resolution the reconstructed sweet-spot bins keep 64–68% purity and their amplitude is within 1–9% of the true-bin value, known through the same response (§7). (2) The single-fill $\cos 2\varphi'$ fit — the estimator of money plots 5–7 — measures the detector's own $\cos 2\varphi'$ acceptance harmonic divided by P_{zz} : a 3% harmonic aligned with the vertical spin axis (the symmetry axis of both the crossing angle and the Roman-Pot cutout) fakes an amplitude four times the signal. The spin-state-sorted ratio of $m = \pm 1$ -rich and $m = 0$ -rich fills cancels any common acceptance bin by bin, turns relative-luminosity errors into a constant, and is 1.5× *more* precise. (3) In the coherent channel the anchored deformation term modulates the *recoil* azimuth $\varphi_t - \varphi_s$, not the electron azimuth used in the projection; the Roman-Pot cutout fakes $\langle \cos 2\varphi_t \rangle \approx 0.5$ for a 25% aspect ratio (against a physics $a_2 \approx 0.036$), and the near-beam cut is angular, so the tag acceptance is 67% / 9% / 10^{-8} at 20.5 / 50 / 137.5 GeV/u for the same optics.

1 • What is measured

Both measurements share the trigger object — a scattered electron in the central detector — and the spin bookkeeping of the ion beam. The coherent measurement adds the intact ⁶Li in the far-forward spectrometers and an exclusivity selection. The observable in both cases is not a cross section but a *Fourier coefficient of a ratio* of spin-sorted azimuthal distributions; Δ follows from that coefficient through kinematic factors and the unpolarized structure functions of the same data set.

RECORDED PER EVENT	INCLUSIVE $E\ ^6\text{Li} \rightarrow E' X$	COHERENT $E\ ^6\text{Li} \rightarrow E' X\ ^6\text{Li}(\text{G.S.})$
Beam and spin	Bunch-crossing ID $\rightarrow$ spin state ($m = \pm 1$ -rich / $m = 0$ -rich pattern); P_{zz} per fill from the polarimeter (scale $\approx 3\%$); alignment axis $\hat{n}$, vertical at IP6, with its measured tilt; luminosity per spin state.	
Central detector	e' candidate: track (p, θ, φ at the vertex) matched to an EMCal cluster, with E/p and shower shape for electron ID; the hadronic final state (all other tracks and clusters) for $\Sigma = \Sigma_h(E_h - p_{z,h})$; vertex.	The same, plus the diffractive system X (its mass M_X and the rapidity gap) — the only handle on x_p , since the recoil momentum is unresolved.
Far forward	Not required (breakup fragments of ⁶ Li are mostly beam-blind at IP6; plans/00).	Roman-Pot track at beam rigidity ($ R - 1 < 0.05$) with transverse displacement above the 10σ envelope (§4 step 1); vetoes in OMD ($p, {}^3\text{He}$), ZDC (n), B0 EMCal (de-excitation γ); Z-identification if available (open question #19).

Observable	$\cos 2\varphi'$ coefficient of the ratio of spin-sorted $N_f(\varphi')$ in (x, Q^2) bins, $\varphi' = \varphi_e - \varphi_S$	$\cos 2\alpha$, $\cos 2\beta$ and $\cos(\alpha+\beta)$ coefficients of the same ratio in (x_p, t, Q^2) bins, $\alpha = \varphi_e - \varphi_S$, $\beta = \varphi_t - \varphi_S$
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2 • Angles and frames

For an unpolarized electron on a spin-1 target whose alignment axis has polar angle θ_S and azimuth φ relative to the virtual photon and the lepton plane, the Hoodbhoy–Jaffe–Manohar cross section for the projection state m reads [1,2]

$$\frac{d\sigma^{(m)}}{dx dy d\phi} = \frac{2y\alpha^2}{Q^2} \left[F_1 + \frac{2}{3} a_m b_1 + \frac{1-y}{xy^2} \left(F_2 + \frac{2}{3} a_m b_2 \right) - \frac{1-y}{y^2} c_m \sin^2 \theta_S \Delta(x, Q^2) \cos 2\phi \right], \quad c_m = 3m^2 - 2$$

and a fill with populations p_m along a transverse axis ($\theta_S = 90^\circ$) has the φ' distribution $1 + a_2 \cos 2\varphi'$ with

$$a_2 = -P_{zz} \frac{1-y}{y^2} \frac{\Delta}{D_\phi}, \quad D_\phi = F_1 + \frac{1-y}{xy^2} F_2, \quad \hat{A} \equiv \frac{a_2}{P_{zz}}, \quad P_{zz} = \sum_m p_m c_m.$$

Which angle. The φ of the master formula is defined in a frame in which the target momentum P and the virtual photon q are collinear; it is the azimuth of the transverse spin about q measured from the lepton plane. The covariant definition used for every transverse-spin azimuth in SIDIS and exclusive analyses [3,4] applies verbatim to the spin four-vector S (and, with P' in place of S , to the recoil azimuth of §4):

$$\cos \phi_S = -\frac{l_\perp \cdot S_\perp}{|l_\perp| |S_\perp|}, \quad \sin \phi_S = -\frac{\epsilon_{\mu\nu\rho\sigma} l^\mu S^\nu P^\rho q^\sigma}{(P \cdot q) \sqrt{1 + \gamma^2} |l_\perp| |S_\perp|}, \quad v_\perp^\mu = g_\perp^{\mu\nu} v_\nu, \quad \gamma^2 = \frac{M^2 Q^2}{(P \cdot q)^2}$$

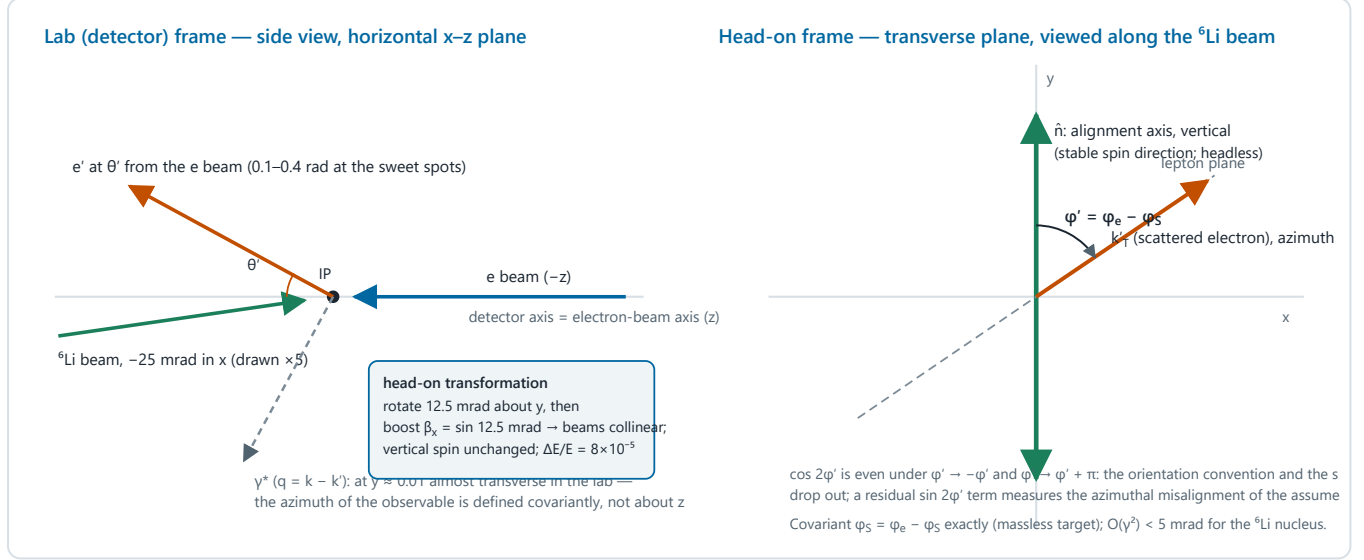
with $g_\perp$ the projector orthogonal to P and q [3]. Evaluated in the head-on frame — ion along $+z$, electron along $-z$, the frame of the whole fast simulation — with a spin axis transverse to the beam at lab azimuth φ_S and the scattered electron at lab azimuth φ_e , this reduces *exactly* to $\varphi' = \varphi_e - \varphi_S$ for a massless target, and to the same with corrections of order γ^2 (below 5 mrad for $\gamma^2 \leq 0.02$, i.e. everywhere in the sweet spots) for the massive nucleus (`reco.azimuth_wrt_lepton_plane`, verified against an explicit boost-and-rotate construction of the collinear frame to 10^{-13}). The lab-angle shortcut is therefore legitimate for the inclusive measurement; the analysis evaluates the covariant formula anyway, because the coherent channel needs the same code.

Which frame. The beams cross at 25 mrad in the horizontal plane at IP6. The ePIC axis coincides with the electron beam, and the ion beam enters at -25 mrad in x (the convention of the far-forward gun scan in `tools/fullsim`). The head-on frame is reached by a rotation of 12.5 mrad about y followed by a boost $\beta_x = \sin 12.5$ mrad; energies change by 8×10^{-5} and the vertical spin direction is untouched. For the scattered electron the boost and the rotation cancel at first order because the detector axis is the electron axis: the residual distortion of its azimuth about the detector axis is an odd-harmonic effect of 0.7×10^{-3} ($\theta' = 107$ mrad) to 2.4×10^{-3} (383 mrad) in $\cos \varphi$, with the $\cos 2\varphi$ harmonic untouched to 10^{-17} and the response of a true $\cos 2\varphi$ term equal to one within 10^{-5} . The transformation is applied, but the crossing angle is not a source of a fake $\cos 2\varphi'$ for the electron. It is one for the recoil (§4).

Orientation and the axis. The tensor alignment axis is headless — the $m = +1$ and $m = -1$ populations enter through $c_m = 3m^2 - 2$ symmetrically — and $\cos 2\varphi'$ is even under $\varphi' \rightarrow -\varphi'$ and $\varphi' \rightarrow \varphi' + \pi$, so neither the orientation convention of φ nor the sign of $\hat{n}$ can change the result. What matters is the axis direction modulo π : a misalignment δ of the assumed azimuth rotates the modulation, $\cos 2(\varphi' - \delta) = \cos 2\delta \cos 2\varphi' + \sin 2\delta \sin 2\varphi'$, so the fit should carry a $\sin 2\varphi'$ term; since the only physics harmonic in the tensor ratio is $\cos 2\varphi'$ (the m -symmetric fills set $a_1 = 0$, and the b_1 -type constant is φ' -independent), the $\sin 2\varphi'$ coefficient measures δ from the data themselves — a self-calibration of the spin axis at the precision of the modulation itself. A polar tilt ε of the axis out of the transverse plane dilutes by $\sin^2 \theta_S = 1 - \varepsilon^2$ (0.8% at 5°).

Harmonics from the tensor structure functions at finite γ . The exact spin-1 decomposition for unpolarized electrons (Cosyn et al. [12], Eq. 10) carries, besides the leading-twist Δ term, tensor-polarized pieces with the same azimuthal signatures: the combination of b_2, b_3, b_4 multiplying $\cos \varphi_{TL}$ is of order γ (their

Eq. 17d), and the combination of b_2, b_3 multiplying $\cos 2\varphi_{TT}$ is of order γ^2 (Eq. 17e), with $\gamma = 2Mx/Q$. At the sweet spots $\gamma^2 \leq 0.023$, so the b-type leakage into the $\cos 2\varphi'$ amplitude is of order $\gamma^2(b_1/F_1)/6 \approx 4 \times 10^{-4}$ for $b_1/F_1 \approx 0.1$ (a generous bound: HERMES' $b_1 \approx 0.11$ at $x = 0.012$ corresponds to $b_1/F_1 \approx 0.01-0.02$) — at most 3% of the Δ amplitude, at the high- x , low- Q^2 corner only, and computable from the same data's A_{zz} ; the $O(\gamma) \cos \varphi'$ term is orthogonal to the $\cos 2\varphi'$ fit at full coverage and becomes a by-product of the ratio fit. (For the axis along q the T_{LT} and T_{TT} parameters vanish, their Eq. 20 — which is why that axis is the clean choice for A_{zz} and the transverse axis the only choice for Δ .) For $x \gtrsim 0.2$ at $Q^2 \approx 1 \text{ GeV}^2$ ($\gamma^2 \approx 0.1$) the leakage grows to tens of percent of a 10^{-3} scenario, so the A_{zz} -based subtraction is required there (docs/code_review_2026-08-25.md, item G2).



Schematic 1 — frames and the azimuth of the observable. Left: the detector frame with the 25 mrad crossing angle; the scattered electron is measured at (E', θ', φ_e) about the electron-beam axis, and the virtual photon is far from the beam axis at low y . Right: in the head-on frame the alignment axis is vertical and the physics azimuth is the angle between the lepton plane and the axis, which the covariant formula reproduces as $\varphi_e - \varphi_S$.

3 • The inclusive chain

- 1. Event selection and electron identification.** One e' candidate with a track matched to an EMCal cluster; E/p and shower-shape cuts against π^- (photoproduction) and a charge-symmetric subtraction (wrong-sign tracks, positron running) for pair-symmetric background at high y . Efficiency $\varepsilon_{eID}(\eta) = 0.70-0.95$ in the ATHENA/ECCE anchors used by the money_delta study; no official ePIC curve exists.
- 2. Scattered-electron kinematics.** E' from the calorimeter ($2\%/\sqrt{E} \oplus 1\%$, i.e. 1.2% at $E' \approx E_e$) or the tracker (3% at $\eta \approx -3$ in the tracking-only table), θ' and φ_e from the track at the vertex (1–3 mrad), transformed to the head-on frame. The four-vectors k, k', P and the spin four-vector $S = (0, \hat{n})$ enter the covariant azimuth.
- 3. Kinematic reconstruction.** The electron method,

$$Q_e^2 = 4E_e E' \sin^2 \frac{\theta'}{2}, \quad y_e = 1 - \frac{E'}{E_e} \cos^2 \frac{\theta'}{2}, \quad x_e = \frac{Q_e^2}{s y_e}, \quad \frac{\delta y_e}{y_e} = \frac{1-y}{y} \left[\frac{\delta E'}{E'} \oplus \tan \frac{\theta'}{2} \delta \theta' \oplus \frac{\delta E_e}{E_e} \right]$$

has a good Q^2 ($\delta Q^2/Q^2 \approx \delta E'/E' \oplus \cot(\theta'/2) \delta \theta'$, dominated by the angle at the sweet spots: 5.2% for the 3 mrad track table used in §7 at $\theta' \approx 0.1 \text{ rad}$, 1.6% at spot 4, 2% if the backward disks reach 1 mrad — the track table is an unsourced placeholder) but loses y as $(1-y)/y$: at $y = 0.01$ a 1.2% energy resolution gives $\delta y/y = 120\%$, and the electron-ring energy spread alone ($\approx 10^{-3}$) gives 10%. Table 1 shows where money plot 5 sits. At the mid-energy configuration three of the four sweet spots are at $y = 0.010-0.025$, the scattered electron carries 98–99% of the beam energy at $\eta \approx -2.9$ to -2.4 , and x is not measurable from e' alone; at the top energy all four spots sit at $y \approx 0.01$. The remedy is the one HERA used at low y [5,6]: y from the hadronic final state,

$$y_{JB} = \frac{\Sigma}{2E_e}, \quad y_{\Sigma} = \frac{\Sigma}{\Sigma + E' (1 - \cos \theta_e)}, \quad \Sigma = \sum_{h \neq e'} (E_h - p_{z,h}), \quad x = \frac{Q_e^2}{s y \Sigma} \text{ (mixed method)}$$

with θ_e measured from the ion direction. The Σ method is insensitive to collinear initial-state radiation, and both are insensitive to particles lost along the ion beam ($E - p_z \approx 0$ for forward remnants); their resolution is set by hadronic calorimetry and by losses in the electron direction. Three documented expectations bracket it: ATHENA uses e- Σ or DA for $y \lesssim 0.1$ and quotes $\delta y/y \approx 25\%$ at $y \approx 0.01$, $\approx 10\%$ at $y \approx 0.1$, and 20–30% for JB throughout (its Sec. 3.1, Fig. 22) [18]; the ePIC kinematic-fit study smears the hadronic $E - p_z$ sum with $\sigma(\delta_h)/\delta_h = 25\%$ and finds $\Delta y/y$ widths of 0.2–0.3 in $0.01 < y < 0.05$, where the electron method is flat [19]; the ATHENA fast simulation of Arratia et al. gives $\text{RMS}(y)/y \approx 0.13\text{--}0.17$ for $\text{I}\Sigma$, DA and JB at $y \approx 0.05\text{--}0.2$ [20]. The band assumed here is 15–30%, with 25% central (the value used in §7). The low-energy configuration moves the same (x, Q^2) to $y = 0.05\text{--}0.25$ (Table 1), so the energy scan is itself part of the reconstruction strategy for the $x \approx 0.1$ bins.

SWEET SPOT	X, Q^2 [GeV ²]	E(5)× ⁶ LI(20.5): Y	E(10)× ⁶ LI(50): Y, E' [GeV], Θ' [MRAD], H	E(18)× ⁶ LI(137.5): Y	$\Delta Y/Y$, E' ALONE (1.2%)
1	0.056, 1.14	0.049	0.010, 9.93, 107, -2.93	0.011 (x = 0.014)	1.2
2	0.022, 1.14	0.124	0.025, 9.77, 108, -2.92	0.026 (x = 0.0056)	0.46
3	0.141, 3.13	0.054	0.011, 9.97, 177, -2.42	0.011 (x = 0.028)	1.1
4	0.141, 14.3	0.247	0.051, 9.85, 383, -1.64	0.010	0.23

Table 1 — the four sweet-spot super-bins of money plot 5 (selected per Q^2 band from the significance map at $e(10)\times^6\text{Li}(50)$) and where the same (x, Q^2) bins sit at the low energy ($y = Q^2/(sx)$ with $s = 4.5\cdot 20.5$ GeV²); the top-energy spots are chosen by the same rule from the top-energy significance map and land at different x ; Figure 1(b) places the re-selected spots of all three energies.

The hadronic final state in the chain. The hadronic y comes from a generated final state: `polligen.hfs` carries the hadronic final state of unpolarized DIS events through a hadron-side detector response and the Σ , Jacquet–Blondel and double-angle methods, and hands the result to the pseudo-experiments as a library — for a pseudo-event at (x, Q^2) the captured fraction of $\Sigma = \Sigma(E - p_z)$ and the p_T ratio of a generator event from the same (x, Q^2) cell are applied to the pseudo-event's exact truth, the calorimeter noise is added per event, and y follows from the chosen method with the *reconstructed* electron. The tensor polarization enters only the inclusive azimuthal weight, so an unpolarized generator is the right source: the production sample is PYTHIA 8 (`tools/pythia8/gen_dis_hfs.py`, $e + p$ and $e + n$ at the per-nucleon beam energies, lepton radiation off), which runs in the eic-shell container; locally the chain runs on a toy string-fragmentation stand-in (`hfs.ToyHFS`), whose numbers are illustrative. The hadron-side response is the detection efficiency the measurement will live with: tracks within $|\eta| \leq 3.5$ above $p_T = 0.2$ GeV (95% plateau), calorimeter objects within $|\eta| \leq 3.7$ (EMCal photons above 0.1 GeV; neutral and untracked charged hadrons in the HCal above 0.5 GeV at 90%), Yellow Report resolution requirements per region, and a Gaussian noise term on Σ and on each p_T component (50 MeV by default, the magnitude Arratia et al. needed to reproduce the H1 full simulation at low y [20]). Figure 3 quantifies the toy: the detector captures 0.90 of Σ (0.60 in tracks, 0.30 in photons, 0.03 in neutral hadrons, the rest below threshold or beyond $|\eta| = 3.7$); without noise the Σ method resolves y to 9–12% everywhere in $0.005 < y < 0.1$, and with 50 MeV of noise the resolution at the four sweet spots is 0.28 / 0.17 / 0.24 / 0.07 (Σ and JB alike; DA 0.37 / 0.29 / 0.28 / 0.10; the electron method 1.2 / 0.46 / 1.1 / 0.23) — the 25% used for §7 is what a 50 MeV noise floor does to a 0.2 GeV $E - p_z$ sum, and 100 MeV would double it. The noise and threshold floor of the ePIC calorimeters at $\Sigma_h \approx 0.2\text{--}0.5$ GeV is therefore the number that decides the $y \approx 0.01$ bins, not the relative resolution at high Σ_h ; the PYTHIA sample replaces the toy's fragmentation, the response parameters stay the same, and both are run by `evgen/scripts/hfs_resolution.py`.

Hadronic-method resolution with a hadronic final state — e(10) x 6Li(50/u); TOY string fragmentation, 396589 events
 tracker $|\eta| \leq 3.5$ pT > 0.20 GeV eff 0.95; cal $|\eta| \leq 3.7$, photon E > 0.10, n-had E > 0.5 eff 0.90; noise 50 MeV

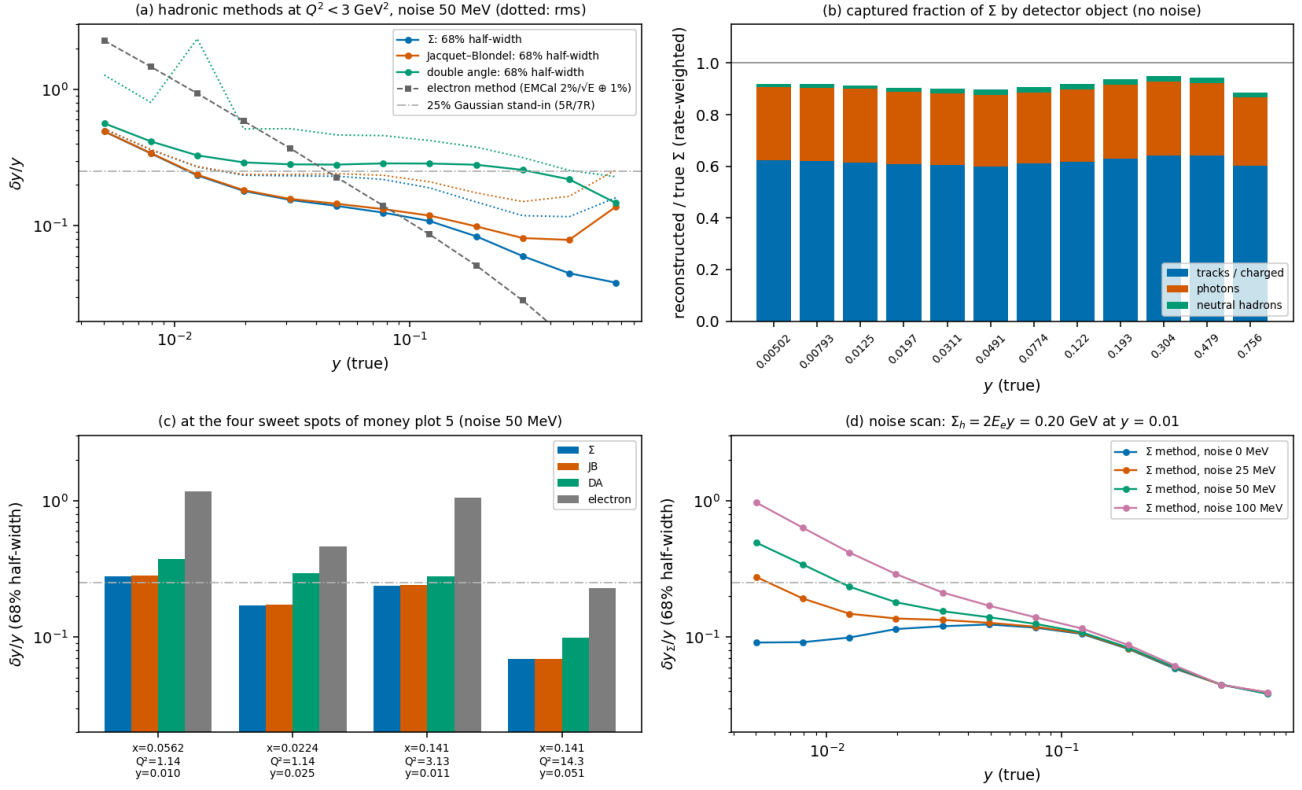


Figure 3 — hadronic-method resolution with a hadronic final state (e10 x $^6\text{Li}50$, toy string fragmentation, `hfs_resolution.py`; illustrative until the PYTHIA 8 sample is generated). (a) $\delta y/y$ (68% half-width; dotted: rms) of the Σ , Jacquet-Blondel and double-angle methods at $Q^2 < 3 \text{ GeV}^2$ with the default response and 50 MeV noise, against the electron method and the 25% stand-in. (b) Captured fraction of Σ by detector object, no noise. (c) The four sweet spots of money plot 5. (d) The Σ -method resolution for calorimeter noise of 0, 25, 50 and 100 MeV: at $y \approx 0.01$ the noise floor, not the fragmentation, sets the resolution.

4. **The azimuth.** $\varphi' = \varphi_e - \varphi_S$ from the covariant formula (§2), with φ_S the alignment-axis azimuth of that bunch from the spin bookkeeping. The track direction resolution enters the azimuth as $\sigma_\varphi = \sigma_\theta / \sin \theta' \approx 20\text{--}30 \text{ mrad}$ at the sweet spots and dilutes $\cos 2\varphi'$ by $\langle \cos 2\delta\varphi' \rangle = 0.985\text{--}0.999$ (§7): negligible.
5. **Binning in reconstructed (x, Q^2).** Figure 1(c) quantifies the migration for the mixed method with the sampler's own events: with $\delta y_{\text{had}}/y = 15\%$ (20%, 30%) the 3×3 super-bins keep a purity of 0.81–0.83 (0.74–0.77, 0.61–0.65) and the amplitude seen in the reconstructed bin is 0.98–0.99 (0.96–0.99, 0.92–0.98) of the true-bin value, because $A(x)$ varies slowly across the bin. With the electron method alone the low- y spots collapse to purities of 0.41–0.56 with 26–53% of the events surviving in the bin. The correction is the standard migration matrix of any structure-function measurement; what is new is only that it must be built on the hadronic-method resolution.
6. **Spin-state sorting and the estimator.** The events of each (x, Q^2) bin are histogrammed in φ' separately for each spin state f of the bunch pattern, $N_f(\varphi')$. Any smooth φ' -dependent efficiency $\varepsilon(\varphi')$ multiplies every N_f identically, so the bin-wise ratio

$$R(\varphi') = \frac{\sum_f w_f N_f(\varphi')}{\sum_f N_f(\varphi')}, \quad w_f = P_f - \bar{P}, \quad \bar{P} = \frac{\sum_f L_f P_f}{\sum_f L_f}, \quad R = \frac{\sigma_P^2 T(\varphi')}{1 + \bar{P} T(\varphi')}, \quad T = \kappa + \hat{A} \cos 2\varphi', \quad \sigma_P^2 = \frac{\sum_f L_f (P_f - \bar{P})^2}{\sum_f L_f}$$

is free of $\varepsilon(\varphi')$ exactly (raw counts with luminosity-weighted w_f , so that $\sum_f L_f w_f = 0$), and a relative-luminosity error δ_f enters it only through $\bar{P}$, as a φ' -independent constant plus a second-order rescaling ($\delta \times \bar{P}/\sigma_P^2$). The cancellation presumes that both spin states see the *same* $\varepsilon(\varphi')$: a difference of its $\cos 2\varphi'$ harmonic between the $m = \pm 1$ -rich and the $m = 0$ -rich samples enters $\hat{A}$ directly as $(\varepsilon_2^+ - \varepsilon_2^0)/(P_+ - P_0) - 10^{-3}$ fakes half the signal —

so the two states must alternate bunch by bunch within a fill (the pattern of Schematic 2), or the harmonic must be stable to 10^{-4} between fills (docs/code_review_2026-08-25.md, §5.1). A weighted least-squares fit of $T(\varphi') = c + d \cos 2\varphi' (+ e \sin 2\varphi')$, with T obtained from R bin by bin (the inversion is exact), gives $\hat{A} = d$ after the finite-bin dilution $\sin w/w$ is divided out, and $\kappa = c$ is the transverse tensor asymmetry of the b_1 sector, a by-product — whose sign follows the program's b_1 transcription rather than the HJM/HERMES convention (Cosyn et al. [12], Eq. 27: $A_{zz} = -(2/3)b_1/F_1$; code review G1), which changes nothing in the Δ sector. For two fill types with equal luminosity the statistical error is

$$\delta\hat{A} = \frac{2}{P_+ - P_0} \sqrt{\frac{2}{N}} \quad (\text{two fills}) \quad \text{vs} \quad \delta\hat{A} = \frac{1}{P_{zz}} \sqrt{\frac{2}{N}} \quad (\text{single fill})$$

so that $m = \pm 1$ -rich bunches at $P_+ = +0.6$ alternated with $m = 0$ -rich bunches at $P_0 = -1.2$ (the same source purity gives $-2P_+$, the pattern of the A_{zz} run plan) are $1.5\times$ more precise than a single $P_{zz} = 0.6$ fill with the whole luminosity; an unpolarized reference fill ($P_0 = 0$) would instead cost a factor two. Figure 1(d) shows the pseudo-experiments: under an efficiency $\varepsilon = 1 + 0.03 \cos 2\varphi' + 0.02 \cos \varphi'$ the single-fill fit returns $\hat{A} = 0.062$ for a true 0.0118 — the acceptance harmonic divided by P_{zz} — while the ratio is unbiased with the analytic error. A percent-level $\cos 2\varphi$ harmonic aligned with the vertical axis is to be expected (the crossing-angle plane is horizontal, module boundaries and dead areas are not azimuthally random, and the Roman-Pot cutout is a rectangle), so the spin-state-sorted ratio is the estimator of the measurement.

7. Amplitude to Δ . With F_2 and $R = \sigma_L/\sigma_T$ of ${}^6\text{Li}$ per nucleon measured in the same data (the unpolarized cross section of the same sample, with its own radiative corrections) or taken from nuclear-PDF fits,

$$\hat{\Delta}(x, Q^2) = -\hat{A} \frac{y^2 D_\phi}{1-y} \rightarrow -\hat{A} \frac{F_2}{x} = -2(1+R)F_1\hat{A} \quad (y \ll 1), \quad \text{i. e.} \quad \frac{\Delta}{F_1} = -2(1+R)\hat{A}$$

At the sweet spots the conversion is therefore insensitive to y — the $(1-y)/y^2$ factor that makes low y attractive for the amplitude cancels against D_ϕ — and Δ/F_1 is essentially a direct ratio measurement; the y mis-measurement of the electron method affects the x label of the point, not the amplitude. Remaining corrections: the tensor rate shift $(2/3)a_m(b_1 + \dots)$ in D_ϕ , at the 10^{-3} level and measured by κ ; target-mass terms of order y^2 ; radiative corrections, uncalculated for tensor observables (plans/07 WP4); the per-nucleon and dilution conventions, which are interpretation rather than reconstruction; and the polarimetry scale $\delta P_{zz}/P_{zz} \approx 3\%$, common with A_{zz} .

Beam and spin bookkeeping

Bunch spin pattern (source)

$m = \pm 1$ -rich bunches, $P_{zz} = +0.6$, interleaved with $m = 0$ -rich bunches, $P_{zz} = -1.2$
→ spin state f per bunch crossing

Polarimetry and spin axis

P_{zz} per fill (scale $\approx 3\%$); axis $\hat{n}$ vertical (stable spin direction); tilt $\rightarrow \sin^2\theta_s$ dilution; axis azimuth self-calibrated by the $\sin 2\varphi'$ term

Luminosity per spin state L_f

bunch-by-bunch; a relative error enters the ratio only as a φ' -independent constant

Dashed arrows: bookkeeping inputs (spin state $\rightarrow$ sorting; P per fill $\rightarrow$ amplitude normalization; $L_f \rightarrow$ yields).

What

ePIC central detector

Scattered-electron candidate

track: p_e, Q_e, φ_e at the vertex, charge; EMCal cluster: E' ($2\%/\sqrt{E} \oplus 1\%$), position; E/p , shower shape $\rightarrow$ e-ID; $\epsilon_{\text{aid}}(\eta) \approx 0.7\text{--}0.95$; charge-symmetric background subtraction

Hadronic final state

all tracks and clusters except e' : $\Sigma \rightarrow \Sigma_H(E_H - p_{zH})$ (robust to forward losses; Σ method also to collinear ISR)

Vertex; bunch-crossing ID; timing

→ spin-state f , beam-gas rejection

Reconstruction and analysis

1 Head-on frame

undo the 25 mrad crossing; four-vectors k, k', P, S (exact; for e' only odd harmonics change, $\approx 10^{-3}$)

2 Kinematic reconstruction

$Q^2 = 4E_e E' \sin^2(\theta'/2)$ (electron, 2–5%)
 $y_2 = Z/(\Sigma + E'(1 - \cos \theta_{\text{lab}}))$ (hadronic, 15–30%)
 $x = Q^2/(s y_2)$ [e' alone: $\delta y/y = 120\%$ at $y = 0.01$]

3 Azimuth

$\varphi' = \varphi_e - \varphi_s$ from the covariant formula (= lab difference to $O(y^2) < 5$ mrad)

4 Histograms per spin state

bin in $(x, Q^2)_{\text{reco}}$; $N_f(\varphi')$ in 24 bins; purity 0.64–0.68 at 25% (0.74–0.83 at 15–20%)

5 Spin-state ratio and fit

$R(\varphi') = \Sigma_f w_f N_f / \Sigma_f N_f, w_f = P_f - \bar{P}$ (L-weighted)
fit $T = c + d \cos 2\varphi' + e \sin 2\varphi'$: common acceptance cancels bin by bin; rel. luminosity $\rightarrow$ constant c only; single-fill fit biased by ϵ 's harmonic $\div P_{zz}$

6 Amplitude and corrections

$\hat{A} = d/\sigma_p^2/\text{dilution}$; $\delta\hat{A} = 2\sqrt{(2/N)/(P_s - P_0)}$
migration matrix / bin centering; radiative corr.; polarimetry scale; axis-tilt dilution

7 Structure function

$\Delta' = -\hat{A} y^2 D_\varphi/(1 - y)$, D_φ from F_2, R of the same data; low y : $\Delta/F_1 = -2(1 + R)\hat{A} \rightarrow x\Delta(x, Q^2)$ points (money plot 7 presentation)

`polligen.reco` + `.hfs` + `.recopseudo` implement: exact frames and angles, Gaussian $E'/\theta'/\varphi_e$ smearing, η -dependent e-ID, a hadronic final state through the hadron-side response, a smooth $\epsilon(\varphi')$, spin-sorted Poisson histograms, the ratio fit (§7). What stays for Phase 2: e-ID purity, radiative events, the real φ acceptance map and its fill-to-fill stability, beam-gas backgrounds.

Schematic 2 — the inclusive reconstruction chain. Left and middle: what is recorded; right: the seven reconstruction and analysis steps of §3 with the quantity each produces and its dominant limitation. The measurement's observable is the $\cos 2\varphi'$ coefficient of the spin-state ratio (step 5), not the modulation of a single histogram.

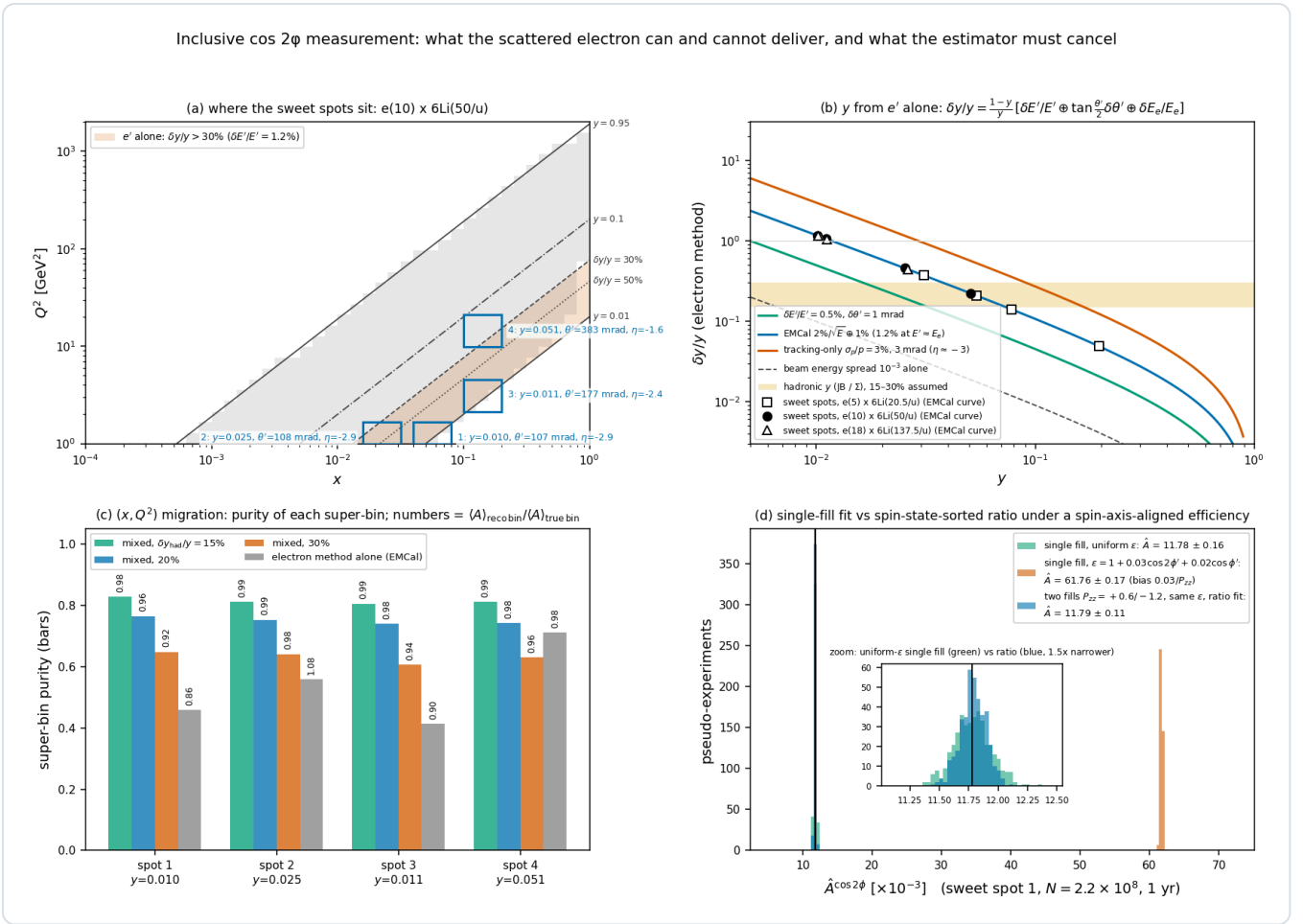


Figure 1 — the inclusive measurement quantified ($e(10) \times 6\text{Li}50$, `reco_chain_figures.py`). **(a)** The analysis acceptance with iso- y lines and the four sweet-spot super-bins of money plot 5; the shaded band is where the electron alone gives $\delta y/y > 30\%$ for a 1.2% energy resolution: spots 1–3 lie inside it, spot 4 just above. **(b)** The electron-method $\delta y/y$ versus y for calorimeter, tracking, and beam-energy-spread inputs, with the sweet spots of the three energies placed on the calorimeter curve and the assumed 15–30% hadronic-method band. **(c)** Purity of each super-bin after smearing the sampler's events with the mixed method for three hadronic- y resolutions and with the electron method alone; the numbers give the ratio of the amplitude averaged over the events landing in the reconstructed bin to that of the true bin. **(d)** Four hundred pseudo-experiments of the spot-1 super-bin at the 1-year statistics: the single-fill fit with a uniform efficiency (green) recovers the truth, the same fit under $\epsilon = 1 + 0.03 \cos 2\phi' + 0.02 \cos \phi'$ (vermillion) is biased by $0.03/P_{zz}$, and the two-fill ratio fit under the same efficiency (blue) is unbiased and $1.5\times$ more precise (inset).

4 • The coherent chain

The coherent measurement keeps steps 1–7 for the electron and adds the recoil. Its kinematics is fixed by two facts of the projection report: the recoil keeps the beam rigidity ($R = 1.000$ for $A/Z = 2$, and the momentum loss $x_p < 0.01$ is below the Roman-Pot momentum resolution), so the only measured recoil quantity is its transverse displacement from the beam orbit; and coherence confines the sample to $x \lesssim 0.01$, i.e. to $y \gtrsim 0.05$ at the mid energy (median $x = 3.5 \times 10^{-3}$, $y \approx 0.2$), where the electron method is adequate and the low- y problem of §3 does not arise.

- Roman-Pot track.** Hits in the two stations (AC-LGAD, ≈ 30 ps timing for bunch association) form a local track; the optics transfer matrices referenced to the beam orbit convert it to the angle pair (θ_x, θ_y) at the IP relative to the ion axis, so the crossing angle and the reference orbit are removed by construction, and the vertex from the central detector supplies the position term. The momentum fraction x_L is set to 1 (unresolved), hence $p_T = A p_u \sqrt{(\theta_x^2 + \theta_y^2)}$ and $\phi_t = \text{atan2}(\theta_y, \theta_x)$. The resolution is dominated by the angular divergence of the bunch itself, $\delta p_T = A p_u \sigma_\theta = 22$ MeV at 50 GeV/u for $\sigma_\theta = 73$ μrad , giving $\sigma(\phi_t) = 0.09$ rad at $p_T = 0.25$ GeV and a $\cos 2\phi_t$ dilution of 1.5% (Figure 2d). The ePIC far-forward simulation with all beam effects (angular divergence, crossing angle, crab rotation and vertex smearing) gives $\Delta p_T/p_T \approx 0.09$ at 0.45 GeV and 0.04 at 1 GeV for 275 GeV protons, i.e. $\Delta p_T \approx 40$ MeV, with the detector alone contributing

$\leq 1.5\%$ — "beam effects the dominant source of momentum smearing" [17]; ZEUS's leading-proton spectrometer was limited the same way ($\sigma(t)/t = 0.14 \text{ GeV}/\sqrt{|t|}$, Φ resolution 0.2 rad) [13].

2. Momentum transfer. Exactly, with $x_L = 1 - x_P$ the plus-momentum fraction,

$$t = -\frac{p_T^2 + M_X^2 x_P^2}{1 - x_P} \simeq -p_T^2, \quad t_{\min} = \frac{M_X^2 x_P^2}{1 - x_P} = 3.2 \times 10^{-3} \text{ GeV}^2 \quad (x_P = 0.01), \quad x_P = \frac{M_X^2 + Q^2 - t}{W^2 + Q^2 - M^2}$$

so the Roman Pots give t , and x_P (or $\beta = x_A/x_P$) must come from the diffractive mass M_X of the central detector (the HERA M_X method [16]) — a coarse variable (a few x_P bins), sufficient for the x_P -dependence test of the two-component fit. The covariant recoil azimuth differs from the lab difference $\varphi_e - \varphi_t$ by less than 0.2 mrad at these kinematics, so here too the lab shortcut is safe.

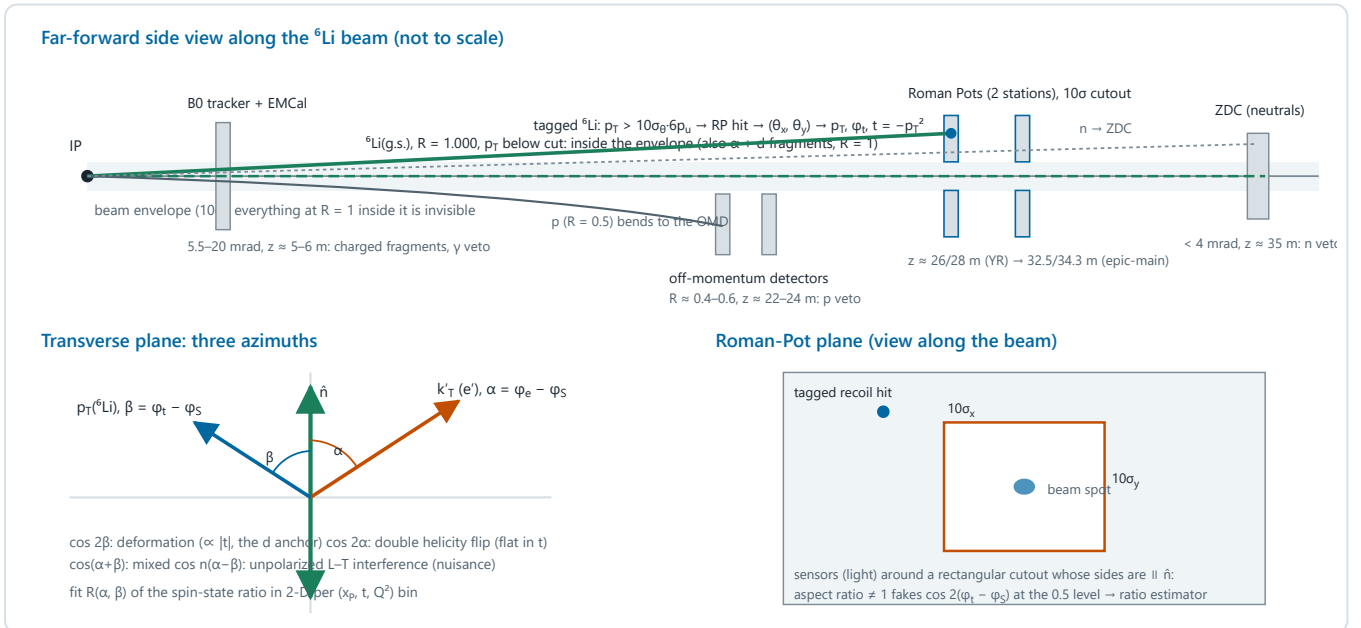
3. Exclusivity selection. A beam-rigidity Roman-Pot track above the envelope; no OMD track (p at $R = 0.5$, ^3He at 0.75 in the main window [8]), no ZDC neutron, no Bo EMCal photon (the 3.56 MeV state's de-excitation γ boosted to $0.1\text{--}0.4 \text{ GeV}$); a rapidity gap between X and the beam; Z -identification of the track if the pots resolve $Z = 3$ from 2 and 1 (open question #19), otherwise a two-component fit of the $|t|$ spectrum against the ten-times-flatter $\alpha + d$ breakup slope; timing and vertex association against beam-gas coincidences. Because the incoherent cross section is m -state-blind, residual breakup dilutes the tensor ratio without faking it; its fraction is measured in the same data from the $|t|$ fit.

4. Three azimuths, two harmonics. With the recoil measured there are two physical azimuths besides the spin: $\alpha = \varphi_e - \varphi_S$ (lepton plane to axis) and $\beta = \varphi_t - \varphi_S$ (recoil to axis), and $\alpha - \beta = \varphi_e - \varphi_t$ is the lepton-plane–recoil angle of unpolarized diffractive DIS. The rank-2 tensor polarization enters only through $e^{\pm 2i\varphi_S}$, so at leading power in the momentum transfer the spin-sorted distribution has the structure

$$N_f(\alpha, \beta) \propto \varepsilon(\alpha, \beta) [1 + u_1 \cos(\alpha - \beta) + u_2 \cos 2(\alpha - \beta) + P_f(\kappa + a_e \cos 2\alpha + a_t \cos 2\beta + a_m \cos(\alpha + \beta))]$$

The deformation term of the polarized-deuteron anchor — the modulation of the diffractive slope by the aligned gluon density, $a_2(t) \propto |t|$, digitized in `coherent.MANTYSAARI_A2_DEUTERON` — is a_t : it lives in the *recoil* azimuth β , requires no photon helicity flip, and vanishes as $t \rightarrow 0$ (the anchor defines Φ as the angle between the vector-meson momentum and the polarization axis in the γ^*d c.m. frame and notes that at $Q^2 > 0$ the transformation from the lab "would mix the polarization states" [9] — the $O(\gamma)$ mixing that the covariant definitions of §2 absorb). The double-helicity-flip term — the coherent counterpart of Δ , an interference of the two transverse photon helicities that needs the linear polarization of the photon and hence the lepton plane — is a_e : it lives in α and is flat in t at leading power. The two mechanisms of the projection report are thus separated not only by their t -dependence but by the azimuth they modulate, which is a stronger discriminant; the mixed term a_m and the unpolarized $u_{1,2}$ are nuisance harmonics. The latter are the L – T and T – T' interference terms of diffractive DIS with a detected recoil, whose y -dependence is $u_1 = (2-y)\sqrt{(1-y)} A_{LT}/[2(1-y)+y^2]$ and $u_2 = 2(1-y) A_{TT}/[2(1-y)+y^2]$ [14]; ZEUS measured them with the leading-proton spectrometer as $A_{LT} = -0.036 \pm 0.036$, $A_{TT} = -0.030 \pm 0.037$ for $x_P < 0.01$ and $+0.051 \pm 0.024$, -0.010 ± 0.024 for $0.01 < x_P < 0.1$ — consistent with zero [13] — and the pQCD LT -interference model gives $A_{LT} \approx 0.03$ at our $\beta \approx 0.5$, $p_T/Q \approx 0.2$ [15]; the pseudo-experiments of §7 use $u_1 = 0.05$, $u_2 = 0.02$, at the edge of the ZEUS bounds. The money-plot-6 pseudo-data place the anchored $\langle a_2 \rangle$ on an axis labelled $\varphi' = \varphi - \varphi_S$ of the electron, which is the α axis: numerically harmless for a statistical projection, but the measurement must histogram (α, β) in two dimensions per (x_P, t, Q^2) bin and extract a_t and a_e from the spin-state ratio $R(\alpha, \beta)$; in that ratio $u(\alpha - \beta)$ survives in the denominator, so the fit must carry it. An *error* δu in the assumed value does not stay second order: it reaches the pure harmonics as $a_t \delta u_2 \langle \cos 2\beta \rangle$ and $a_e \delta u_1$, and with the cutout's fake $\langle \cos 2\beta \rangle = +0.77$ and $a_t/a_e \approx 12\text{--}34$ the first of these dominates — a ZEUS- 1σ $\delta u_2 = 0.024$ shifts a_e by 20%, 1.6 one-year statistical errors, while the same δu_1 moves a_e by under 1% and shows up in the mixed term instead (§7.1). u_2 , not u_1 , is the nuisance to measure in situ from the spin-averaged counts of the same data.

- 5. Acceptance geometry and the estimator.** The Roman-Pot acceptance is not a circular p_T cut. The ePIC pots are 25.6×12.8 cm AC-LGAD planes surrounding a *horizontal slot* through which the beam, its momentum spread and its dispersion pass [17]: the cutout is wide in x and tight in y (the 10σ vertical beam size), a rectangle with $c_x \gg c_y$ whose sides are again horizontal and vertical — parallel and perpendicular to the spin axis. Anisotropy is the norm, not the exception: HERA's proton beam had a transverse-momentum spread at the interaction point of 45 MeV horizontally and 100 MeV vertically [13]. For the exponential recoil spectrum the tagged sample's azimuthal distribution is $\varepsilon(\varphi_t) = \exp[-B \rho(\varphi_t)^2]$ with ρ the cutout boundary, and its $\langle \cos 2\varphi_t \rangle$ (about the horizontal) is 0 for a square, +0.49 for an aspect ratio $c_y/c_x = 1.25$, +0.71 for 1.5, and -0.36 to -0.45 for the slot-like ratios 0.3–0.6 of Figure 2b (-0.77 for the wider slot of §7, 0.55×0.22 GeV) — ten to twenty times the anchored $a_t \approx 0.036$, with the same symmetry axis (the sign flips when the axis is referred to the vertical spin direction). A square still carries $\langle \cos 4\varphi_t \rangle = 0.34$, which mixes into $\cos 2\beta$ at second order through the physics term. No single-fill fit can survive this; the spin-state ratio of §3 cancels it exactly, bin by bin in (α, β) , and the same $m = \pm 1$ -rich / $m = 0$ -rich pattern serves both measurements.
- 6. The near-beam cut is angular.** The documented 0.20 GeV (high-acceptance) and ≈ 0.41 – 0.45 GeV (high-divergence) cuts are ten beam divergences times the 275 GeV proton momentum [7], $\sigma_\theta \approx 73$ and 149 μrad . The same envelope applied to a ${}^6\text{Li}$ nucleus of momentum $6p_u$ is $p_T^{\text{cut}} = 10\sigma_\theta \cdot 6p_u$: 0.09, 0.22 and 0.60 GeV at 20.5, 50 and 137.5 GeV/u, giving tag acceptances of 67%, 9% and 1.5×10^{-8} for $B = 50 \text{ GeV}^{-2}$ (Figure 2c). The projection's constant 0.20 GeV cut (13.5%) is close to the mid-energy value and far from the others; whether the lithium divergence at a given energy equals the proton's is precisely the undocumented optics question of plans/04 #11, but the scaling with $A p_u$ is kinematics, not optics, and it makes the low-energy configuration the natural home of the coherent program (the alternative being the IR-8 secondary focus, which reaches $p_T \approx 0$ [10]).
- 7. Resolution and unfolding.** The divergence smearing moves the effective slope from B to $1/(1/B + 2\delta p_T^2)$ (47.7 for the numbers above) and smears $|t|$ by $2p_T\delta p_T \approx 0.01 \text{ GeV}^2$ — small against the 0.04–0.2 GeV^2 window (Figure 2d); the t -binning migration and the acceptance edge are unfolded with the same emulation, and $t_{\min}$ is a known bias of $3 \times 10^{-3} \text{ GeV}^2$ at $x_p = 0.01$ once x_p is known from M_X .



Schematic 3 — the coherent channel. Top: the far-forward measurement of the intact recoil, visible only when its transverse momentum carries it outside the beam envelope at the Roman Pots, and the vetoes that define exclusivity. Bottom left: the two physical azimuths relative to the alignment axis and the harmonic each mechanism modulates. Bottom right: the rectangular cutout of the pots, whose symmetry axes coincide with the spin axis.

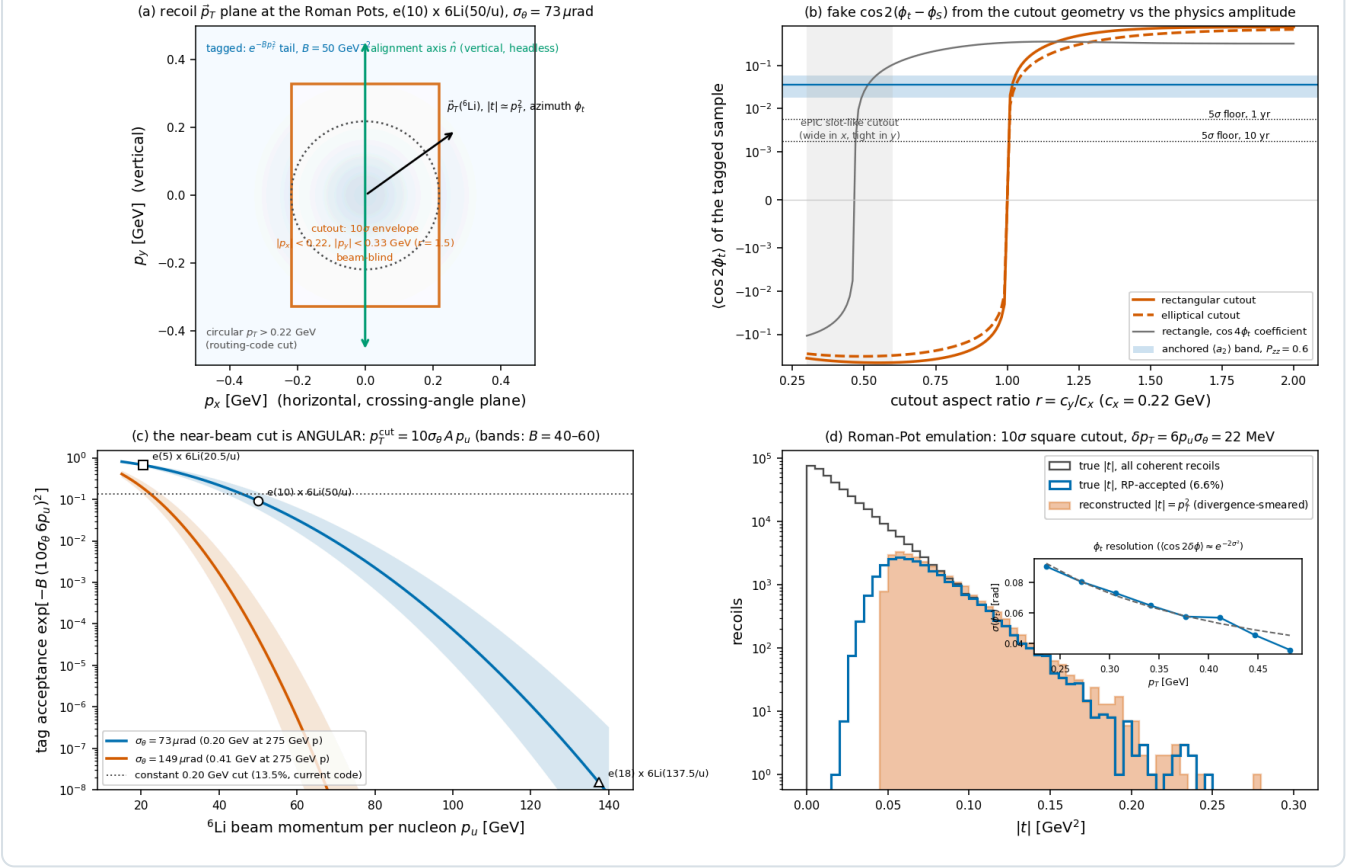


Figure 2 — the coherent measurement quantified. (a) The recoil transverse-momentum plane: the exponential coherent spectrum, the rectangular 10σ cutout (aspect ratio 1.5 drawn) against the circular cut of the routing code, and the vertical alignment axis. (b) $(\cos 2\phi_t)$ of the tagged sample versus the cutout aspect ratio for rectangular and elliptical cutouts, with the $\cos 4\phi_t$ coefficient of the square, the anchored deformation band and the 5σ statistical floors of the 1- and 10-year programs. (c) Tag acceptance versus beam momentum per nucleon for the two proton-derived divergences, with the $B = 40-60$ GeV⁻² bands, the three beam configurations, and the constant-cut value of the projection. (d) The Roman-Pot emulation (`reco_rp_measure`): true and accepted $|t|$ spectra, the reconstructed $|t| = p_T^2$ after divergence smearing, and (inset) the ϕ_t resolution against $\delta p_T/p_T$.

5 • What the simulation does

The table walks the chain of §§3–4 and states, for each step, what an experiment does and what the repository [11] does today. The generator kernel and the spin bookkeeping were complete from the start; the measurement layer — four-vectors, frames, kinematic reconstruction, the hadronic final state, the reconstructed azimuth, the spin-state ratio, the Roman-Pot measurement — is `polligen/reco.py` and `polligen/hfs.py`, composed by `polligen/recopseudo.py` into the pseudo-experiments of §7. The column worth reading is the last one.

STEP	EXPERIMENT	REPOSITORY	STATUS
Spin bookkeeping, luminosity per state, polarimetry	Bunch pattern, L_f , P_{zz} per fill with a 3% scale	<code>bookkeeping.SpinCategory</code> / <code>RunPlan</code> / <code>tensor_flip_plan</code> : categories with lumi shares, relative-luminosity offsets, polarimetry smearing	done
e' four-vector, crossing angle, head-on frame	Track + cluster $\rightarrow k'$ in the lab; head-on transformation	<code>reco.electron_fourvector</code> , <code>head_on_to_lab</code> , <code>lab_to_head_on</code> , applied per event by <code>recopseudo.RecoResponse</code>	done
e' resolution	Calorimeter energy, track angles	<code>reco.emcal_resolution</code> , <code>tracking_resolution</code> , <code>tracking_angular_resolution</code> , <code>smear_electron</code> . The EMCal term $2\%/\sqrt{E} \oplus 1\%$ is the backward specification applied at every η ; the angular table has no published source	done, placeholder numbers

Electron ID and efficiency	E/p, shower shape, charge-symmetric subtraction	<code>reco.eps_eid(η)</code> on the ATHENA/ECCE anchors; no purity or background model	partial
Kinematic reconstruction (electron / JB / Σ / DA / mixed)	Mixed method at low y from the hadronic final state	<code>reco.electron_method</code> , <code>mixed_method</code> ; <code>hfs.hadronic_kinematics</code> for Σ , JB and DA on a generated final state, or the parametrized <code>reco.hadronic_y</code>	done
Hadronic final state and hadron-side detection	All tracks and clusters except e' , through the detector	<code>hfs.HadronResponse</code> (tracker coverage and thresholds, EMCal/HCal, calorimeter noise) on <code>hfs.ToyHFS</code> ; the PYTHIA 8 sample of <code>tools/pythia8</code> needs eic-shell	done with a toy generator
The physics azimuth	Covariant φ_S from k, k', p, S	<code>reco.azimuth_wrt_lepton_plane</code> , verified against an explicit collinear-frame construction to 10^{-13}	done
Reconstructed (x, Q^2) binning and migration	Reco bins, migration matrix, unfolding	<code>RecoResponse.mask_reco</code> / <code>bin_summary</code> : purity, efficiency and the reco-bin amplitude per bin; the correction is an MC bin-centering factor, not an unfolding	done, no unfolding
Spin-state-sorted ratio estimator	$R(\varphi')$ of $m = \pm 1$ -rich vs $m = 0$ -rich fills; fit with $\sin 2\varphi'$	<code>reco.spin_state_ratio</code> , <code>harmonic_ratio_fit</code> and its two-azimuth form, with the fill-dependent efficiency the ratio cannot cancel (<code>fill_acceptance_bias</code> , <code>money_cos2phi_reco.py --eff-cos2-split</code>)	done
Full-luminosity pseudo-data	—	<code>sample.phi_histogram_pseudo</code> (truth level) and <code>RecoResponse.expected_counts</code> (reconstructed level): exact expected counts per φ' bin, Poisson-fluctuated	done
Amplitude $\rightarrow \Delta$	Kinematic conversion with measured F_2, R ; unfolding	<code>recopseudo.delta_from_amplitude</code> : model bin-centering factor K with migration included; F_2 from the model rather than from the same data	partial
Radiative corrections	Unpolarized RC on the yields; tensor RC	None (plans/07 WP4 bound planned)	missing
Coherent: recoil four-vector and t	t from the recoil, x_p from M_X	<code>reco.recoil_fourvector</code> (exact light-cone), <code>t_from_fourvectors</code> ; x_p is sampled, never reconstructed — M_X and the diffractive system are absent	partial
Coherent: Roman-Pot measurement	Transport, divergence smearing, cutout geometry, angular cut	<code>reco.rp_measure</code> , <code>rp_hole_acceptance</code> , <code>tag_pt_cut = 10\sigma_0 6p_{U'}</code> ; <code>coherent.py</code> keeps the older scalar p_T cut for the truth-level projections	done
Coherent: two-azimuth harmonic analysis	$R(\alpha, \beta)$ per (x_p, t, Q^2) bin	<code>reco.harmonic_ratio_fit_2d</code> with the acceptance-weighted template basis (§7), and the assumed u_1, u_2 , luminosity shares and per-fill cutout as fit arguments; α is integrated analytically — there is no reconstructed electron on the coherent side	done, no electron side
Coherent: exclusivity and vetoes	OMD/ZDC/B0 vetoes, rapidity gap, Z-ID, $ t $ fit	<code>coherent.veto_table</code> bookkeeping only; no event-level breakup model (BeAGLE blocked on FLUKA)	missing
Full-simulation far-forward	Geant4 acceptance, EICrecon transport	<code>tools/fullsim</code> gun scans (hit level, RP at 32.5/34.3 m); no ${}^6\text{Li}$ -nucleus gun, no reconstruction	partial

Verdict. The chain is closed end to end at the reconstructed level, and three of its steps changed the headline numbers rather than their decoration: the hadronic-method requirement at the $y \approx 0.01$ sweet spots (which changes the binning and the energy strategy, not the reach), the estimator ($0.67\times$ the error and one uncontrolled systematic removed), and the coherent angular cut together with the cutout geometry. What is still absent is everything downstream of the response — radiative corrections, an event-level breakup background, x_p from M_X , and an unfolding in place of the model bin-centering factor.

6 • What remains

The six recommendations of the first version of this note are carried out (§7 and Appendix A) except in three respects, which are the open items of the measurement design.

1. **The hadronic-resolution band.** 5R and 7R are quoted at the 25% central value; the 15–30% band and the low-energy configuration for the $x \approx 0.1$ bins are not yet run. The one number the analysis cannot produce for itself is the calorimeter noise and threshold floor at $\Sigma_h \approx 0.2\text{--}0.5$ GeV, which is what decides the $y \approx 0.01$ bins (plans/04 #21); the chain that turns it into a $\delta y/y$ exists (§3).
2. **The coherent optics.** The angular cut is in place, but the acceptance, yield and δa_t curves versus σ_θ and p_u are plans/07 WP5, and the Roman-Pot cutout dimensions for the light-ion optics remain an explicit assumption (plans/04 #20).
3. **Everything downstream of the response.** Radiative corrections (WP4); an event-level $\alpha + d$ breakup background with the two-component $|t|$ fit (plans/06); x_p from the diffractive mass; an unfolding in place of the model bin-centering factor; and a fill-dependent ϕ' efficiency, the one systematic the ratio estimator cannot cancel (§7.1).

7 • Reconstructed-level closure: money plots 5R, 7R and 6R

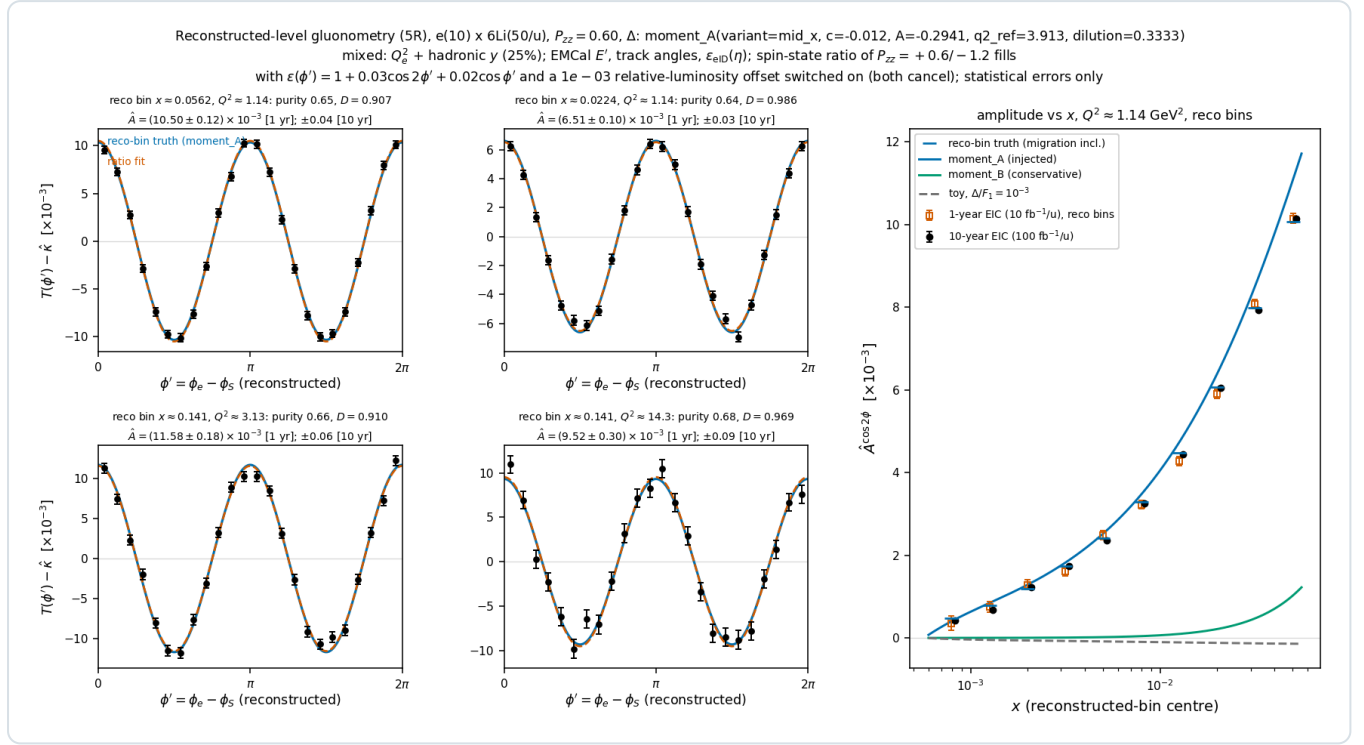
The chain of §§3–4 is composed in `polligen/recopseudo.py`. The inclusive response is importance-sampled: every accepted sampler cell (60×45 grid, generator cuts loosened to $y \geq 0.004$, $Q^2 \geq 0.7$ GeV², $|\eta| \leq 3.8$, $E' \geq 0.3$ GeV so that events can migrate *into* the analysis window) receives 400 pseudo-events with weight σ_c/n_{kept} (n_{kept} = those inside the generator acceptance, so the cell rate is conserved), each carried through the lab frame (25 mrad crossing), Gaussian smearing of E' (EMCal 2%/√E ⊕ 1%) and of the track direction (1–5 mrad by η), the head-on transformation, the electron-method Q^2 , a hadronic y with the documented 25% resolution [18,19], the mixed-method x , the analysis cuts on reconstructed quantities, $\varepsilon_{\text{eID}}(\eta)$, and the covariant ϕ' on true and smeared four-vectors. The expected ϕ' counts of any reconstructed bin are then exact bin integrals of $\text{den}_f + P_f A \cos 2\phi'$ summed over the events that land in it (the reconstructed-level analogue of `phi_histogram_pseudo`), Poisson- fluctuated at the 1-year and 10-year luminosities for the two fills of the $m = \pm 1$ -rich / $m = 0$ -rich pattern (`bookkeeping.tensor_flip_plan`), with the spin-axis-aligned efficiency $\varepsilon(\phi') = 1 + 0.03 \cos 2\phi' + 0.02 \cos \phi'$ and a 10^{-3} relative-luminosity offset switched on. The estimator is the spin-state ratio fit of §3 with the $\sin 2\phi'$ term; Δ follows from the amplitude with the MC bin-centering factor $K = \Delta_{\text{model}}(x_c, Q_c^2)/A_{\text{model, reco bin}}$, which includes the migration.

SWEET SPOT (RECO BIN)	N, 1 YR (BOTH FILLS)	PURITY	EFFICIENCY	D = A _{RECO} BIN/A _{TRUE} BIN	$\hat{A} \pm \Delta \hat{A}$ (1 YR) [10 ⁻³]	RECO-BIN TRUTH	$\Delta \hat{A}$ (10 YR)	ΔA OF MONEY PLOT 5
1: $x = 0.056$, $Q^2 = 1.14$	1.7×10^8	0.65	0.42	0.907	10.50 ± 0.12	10.34	0.038	0.18
2: $x = 0.022$, $Q^2 = 1.14$	2.8×10^8	0.64	0.60	0.986	6.51 ± 0.10	6.60	0.030	0.15
3: $x = 0.141$, $Q^2 = 3.13$	7.6×10^7	0.66	0.37	0.910	11.58 ± 0.18	11.71	0.058	0.26
4: $x = 0.141$, $Q^2 = 14.3$	2.8×10^7	0.68	0.64	0.969	9.52 ± 0.30	9.32	0.095	0.45

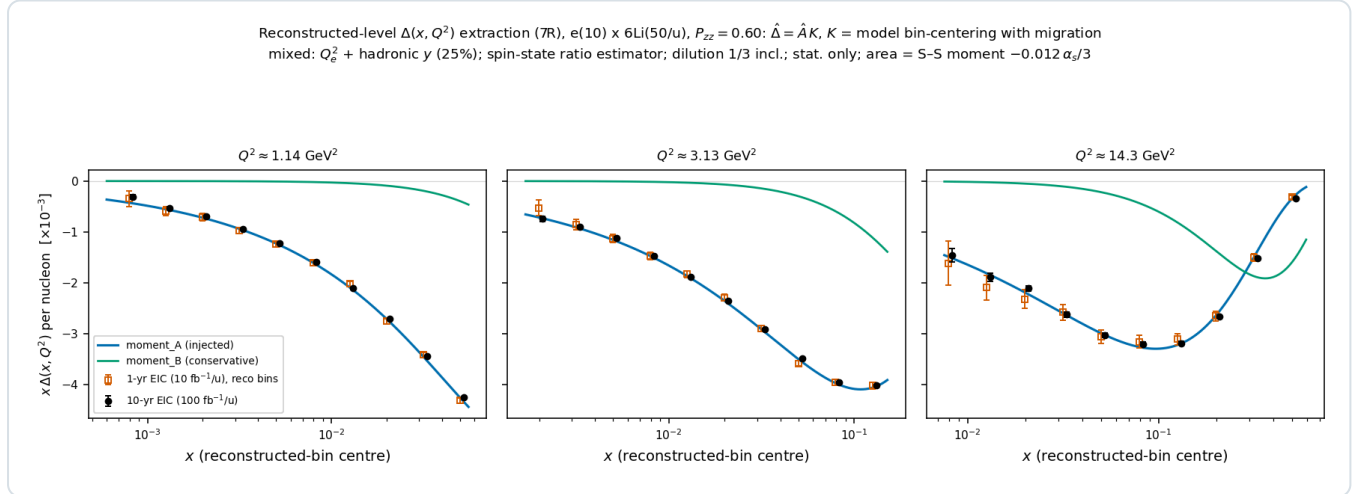
Table 2 — money plot 5R: the four sweet-spot super-bins in reconstructed (x, Q^2) with the mixed method and the documented 25% hadronic- y resolution [18,19]. Efficiency = fraction of the true-bin rate reconstructed in the bin after ε_{eID} ; D = ratio of the rate- and ε_{eID} -weighted, ϕ' -diluted model amplitude of the events in the reconstructed bin to the rate-weighted amplitude of the true bin. The last column is the single-fill, truth-binned error of the projection report.

Three observations. (i) The fit is unbiased at the reconstructed level: every $\hat{A}$ agrees with its reco-bin truth within the statistical error while the acceptance harmonic and the luminosity offset are on, and the $\sin 2\phi'$ coefficients are consistent with zero. (ii) The two edge bins (spots 1 and 3 at $y \approx 0.01$) lose about 60% of their events to the reconstructed $y \geq 0.01$ cut and their amplitude reference shifts by 9% (the accepted half of the bin sits at lower true x); the analysis knows both through K , and the moment-constrained model's x -dependence is still resolved bin by bin (money plot 7R, best bins $\delta \Delta = 1.0 \times 10^{-3}$ at $Q^2 = 1.14$ and 0.5×10^{-3} at 3.13 GeV² in year

1, purities ≈ 0.55). (iii) The errors are *smaller* than the truth-binned single-fill projection — 0.64–0.70 of the money-plot-5 values — because the $1.5\times$ gain of the $m = 0$ -rich fill outweighs the reconstruction efficiency of 0.37–0.64; this is the number to carry into plans/o7 §7.1, with the caveat that it presumes the source delivers $m = 0$ -rich bunches at $|P_{zz}| = 1.2$ (the same purity as the ± 1 fill at 0.6).



Money plot 5R — inclusive gluonometry at the reconstructed level ($e(10) \times 6\text{Li}(50/u)$, $P_{zz} = 0.6$, moment_A, mixed method with 25% hadronic y , EMCal energy, track angles, $\epsilon_{\text{elD}}(\eta)$, spin-state ratio of $P_{zz} = +0.6/-1.2$ fills with the acceptance harmonic and luminosity offset on). Left: the acceptance-free modulation $T(\phi') - \kappâ$ (per unit P_{zz}) from the ratio, with the reco-bin truth (blue) and the fit (vermillion); annotations give purity, D , and the 1-/10-year errors. Right: the amplitude in reconstructed x bins along the $Q^2 = 1.14 \text{ GeV}^2$ slice with the reco-bin truth (blue ticks) and the model curves.



Money plot 7R — $\Delta(x, Q^2)$ from reconstructed bins. $x\Delta$ points at the three sweet-spot Q^2 slices from the same pseudo-measurements, converted with the MC bin-centering factor K (model-derived, as in money plot 7, so the points sit on the injected curve by construction and carry the statistical errors of the reconstructed analysis); 1-year (open) and 10-year (filled) draws are independent.

Cross-check with the hadronic final state. Rerunning 5R with the hadronic y from the toy final state through the hadron-side response (Σ method, 50 MeV noise; `money_cos2phi_reco.py --y-source hfs`) instead of the 25% Gaussian gives purities 0.60 / 0.68 / 0.68 / 0.86, efficiencies 0.41 / 0.65 / 0.39 / 0.84, $D = 0.900$ / 0.939 / 0.913 / 0.986 and $\delta\hat{A} = 1.2$ / 1.0 / 1.8 / 3.0×10^{-4} at the four spots — the same errors as Table 2, a worse spot 1 and a better spot 4 (where $y = 0.05$ and the resolution is 7% rather than 25%), with the fit unbiased and the median captured Σ fraction 0.99 for the selected events; 7R is unchanged ($\delta\Delta = 1.0 \times 10^{-3}$ at $Q^2 = 1.14$ and

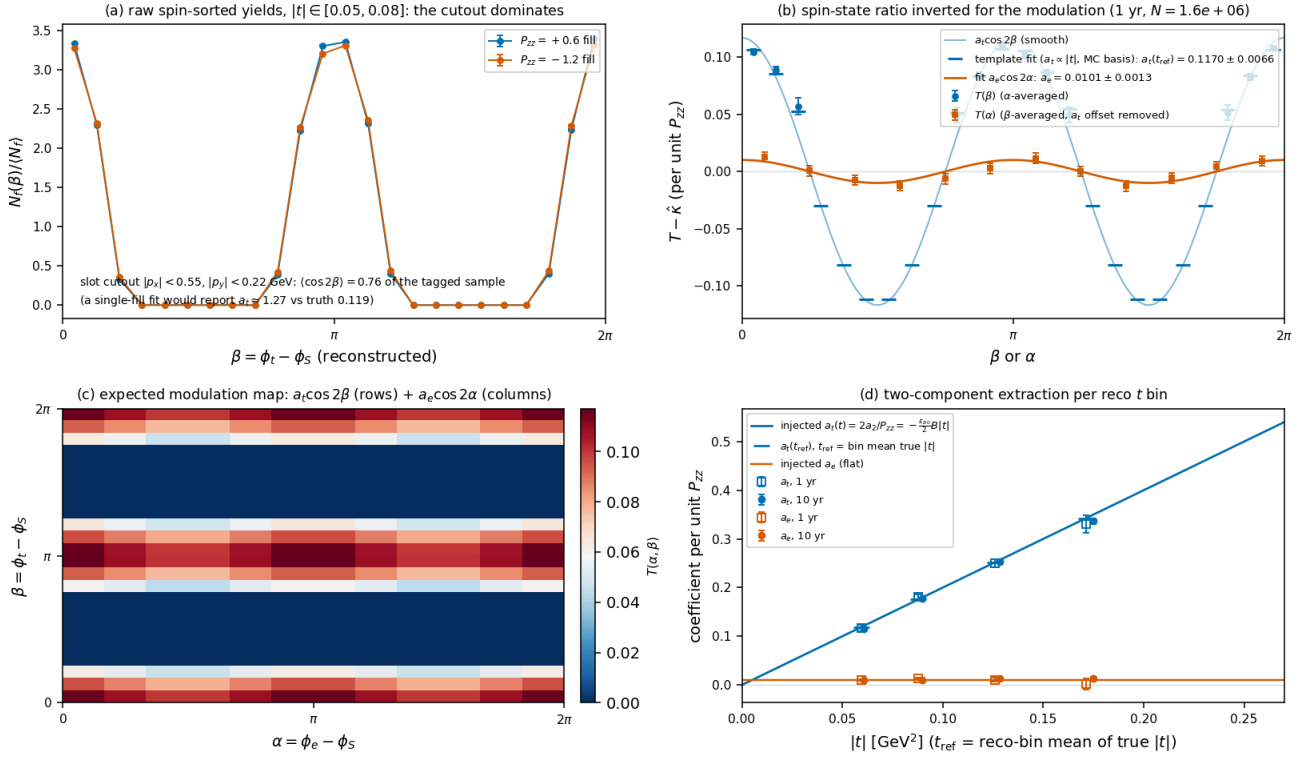
0.5×10^{-3} at 3.13 GeV^2 in year 1). The Table 2 numbers therefore stand as long as the calorimeter noise floor is $\approx 50 \text{ MeV}$; they degrade in proportion to it.

Coherent channel (money plot 6R). The coherent response samples recoils from the exponential spectrum with $x_p \in [10^{-3}, 10^{-2}]$, builds the exact four-vector, and emulates the pots with the angular envelope $10\sigma_0 \cdot 6p_u$ at $\sigma_0 = 73 \text{ } \mu\text{rad}$ (0.22 GeV at 50 GeV/u), a slot-like cutout $|p_x| < 0.55$, $|p_y| < 0.22 \text{ GeV}$ (the horizontal slot of the ePIC sensor planes [17], its width illustrative), and divergence smearing. The two-azimuth expected counts per fill are exact α -integrals per event of $1 + u_1 \cos(\alpha - \beta) + u_2 \cos 2(\alpha - \beta) + P_f[a_e \cos 2\alpha + a_t(t) \cos 2\beta + a_m \cos(\alpha + \beta)]$ with the *true* β and t inside and the reconstructed β and $|t| = p_T^2$ for the binning. Two lessons follow. First, the anchor convention: the full text of the deuteron paper fixes $d^2\sigma/d\Phi d|t| \propto 1 + 2\Sigma_n a_n e^{in\Phi}$ (its Eq. 9, with Φ the angle between the vector-meson momentum and the polarization axis in the γ^*d frame), so the $\cos 2\beta$ coefficient is $2a_2$ — the deformation modulation is twice what money plot 6 injects ($c_2 = -(P_{zz}/2)\epsilon_{Bo}B|t| = 0.072$ at $\langle|t|\rangle = 0.06 \text{ GeV}^2$ and $P_{zz} = 0.6$; `coherent.cos2phi_coefficient_deformation`). Second, the cutout makes the in-bin β distribution far from uniform (the acceptance varies by more than an order of magnitude around the circle) and correlates β with $|t|$ inside a reco t bin (the blind directions hold the larger- $|t|$ recoils), so the harmonic fit must use the acceptance-weighted basis from the response — $\langle \cos 2\beta \rangle$, $\langle \cos \beta \rangle$, $\langle \sin \beta \rangle$ per bin, and for the t -dependent term the template $\langle (t/t_{\text{ref}}) \cos 2\beta \rangle$ of the assumed linear shape — which is exactly the two-component fit $a_2(t) = c_{\text{def}}|t| + a_g$ of plans/06 in MC-response form. With the analytic (uniform-bin) basis the deformation coefficient is biased by 3–6% in the lowest t bin (1.25-aspect cutout); with the template it closes. A third, geometric lesson: the slot concentrates the tagged recoils near $\beta \approx 0$ and π , where $\cos 2\beta$ hardly varies, so the lever arm on the deformation harmonic shrinks — its statistical error is three times that of a square cutout of similar acceptance (0.007 against 0.002 in the first $|t|$ bin at one year), while the electron-azimuth term a_e is untouched. The pot geometry is therefore a design lever for the deformation measurement and irrelevant for the exotic-gluon one.

RECO $ T $ BIN $[\text{GeV}^2]$	$N_{\text{TAG}}, 1 \text{ YR}$	$A_T(T_{\text{REF}})$ INJECTED	FIT, 1 YR	FIT, 10 YR	A_E FIT, 1 YR (INJ. 0.010)	A_E FIT, 10 YR
0.05–0.08	1.6×10^6	0.119	0.117 ± 0.007	0.1161 ± 0.0021	0.0101 ± 0.0013	0.0100 ± 0.0004
0.08–0.12	7.4×10^5	0.176	0.181 ± 0.005	0.1772 ± 0.0014	0.0130 ± 0.0018	0.0095 ± 0.0006
0.12–0.17	1.3×10^5	0.252	0.250 ± 0.008	0.2533 ± 0.0024	0.0106 ± 0.0043	0.0137 ± 0.0014
0.17–0.25	1.8×10^4	0.343	0.331 ± 0.018	0.337 ± 0.006	0.003 ± 0.012	0.013 ± 0.004

Table 3 — money plot 6R: template fit of the two-azimuth spin-state ratio per reconstructed $|t|$ bin (coefficients per unit P_{zz} ; $\epsilon_{Bo} = -0.08$, $B = 50 \text{ GeV}^{-2}$; $u_1 = 0.05$, $u_2 = 0.02$ at the ZEUS 1σ bounds [13]; $N_{\text{tag}} = 2.7 \times 10^6$ in year 1 for the angular envelope and the slot-like cutout, against 1.1×10^7 for the constant 0.20 GeV cut). The cutout's fake $\langle \cos 2\beta \rangle$ of the tagged sample is $+0.77$ (a single-fill fit would report $a_t \approx +1.3$ for a true 0.12); the ratio removes it entirely. The sample is background-free and Z-identified by assumption (plans/06 carries the breakup budget).

Coherent $e^6\text{Li} \rightarrow e^+X^6\text{Li}(\text{g.s.})$ at the reconstructed level (6R), $e(10) \times 6\text{Li}(50/\text{u})$, $P_{zz} = 0.60$: $\sigma_\theta = 73 \mu\text{rad}$ ($10\sigma_\theta A_{p_u} = 0.22 \text{ GeV}$), slot-like rectangle cutout $|p_x| < 0.55$, $|p_y| < 0.22 \text{ GeV}$
 $N_{\text{tag}} = 2.7\text{e}+06$ (1 yr) vs $1.1\text{e}+07$ with the constant 0.20 GeV cut; deformation $c_2 = 2a_2$ (Eq. 9 convention), $a_e = 0.010$
 $u_1 = 0.05$, $u_2 = 0.02$ (ZEUS LPS 1σ bounds); two-fill ratio fit with the MC template basis; statistical errors only



Money plot 6R — the coherent channel at the reconstructed level. (a) Raw spin-sorted yields versus the reconstructed recoil azimuth β for the lowest $|t|$ bin: the slot-like cutout shape dominates both fills identically (vertical recoils, $\beta \approx 0$ and π , are the ones that clear the slot). (b) The modulation $T(\beta)$ and $T(\alpha)$ recovered from the spin-state ratio ($T(\alpha)$ with the a_t offset removed), the template-fit prediction per β bin (ticks) and the smooth $a_t \cos 2\beta$ and $a_e \cos 2\alpha$ curves. (c) The expected (noise-free) modulation map in (α, β) : rows from the deformation term, columns from the gluon-transversity term. (d) Extracted $a_t(t_{\text{ref}})$ and a_e per reco $|t|$ bin (1 yr open, 10 yr filled) against the injected curves.

7.1 • Audit notes and known limits

Checked independently of the code that produced the numbers: the analytic error of the two-fill ratio, $2\sqrt{(2/N)/(P_+ - P_-)}/\text{dilution}$, reproduces the pseudo-experiment spreads and the printed errors (spot 1: 1.21×10^{-4} for $N = 1.72 \times 10^8$); the exact expected counts reproduce the sampler's generator-level `effective_modulation` when binned in truth (test); the bin-centering factor returns the injected Δ exactly without noise (test); the sign of the deformation term is positive for $P_{zz} > 0$ and $\varepsilon_{B0} < 0$, consistent with the predicted sign flip relative to the deuteron; the Eq. (9) convention was read from the paper's text, not inferred; the relative-luminosity immunity is second order ($\delta \cdot P/\sigma_P^2$ — a 2% offset moves the amplitude by 0.7%, a 10^{-4} offset by 3×10^{-5}); empty ϕ bins inside the cutout are handled with zero weight; the α -integrals of the coherent counts are analytic per event and the β smearing is absorbed in the MC basis, so the fitted coefficients refer to the unsmeared truth.

Known limits, stated rather than hidden. The hadronic y of Table 2 is a parametrized 25% Gaussian — an absolute 50–125 MeV on $\Sigma_h = 0.2\text{--}0.5 \text{ GeV}$ at the sweet spots — reproduced by the toy final state with a 50 MeV noise floor (§3), so the open item is the noise and threshold floor of the real detector at that Σ_h and the PYTHIA fragmentation in place of the toy (plans/04 #21); u_1, u_2 are generated at the ZEUS 1σ bounds [13] and the slot dimensions are illustrative (#20); the ϕ' -efficiency harmonics are illustrative magnitudes; radiative corrections and backgrounds are not included; the 7R points are centered with the injected model (as in money plot 7); the template assumes the linear t-shape it fits, so a different shape is a model systematic to be varied. From the code review (docs/code_review_2026-08-25.md): the ϕ' efficiency of Table 2 is common to both fills, and the ratio

cancels only a common acceptance. The fill-dependent case is now modelled rather than argued — a difference $\delta\epsilon_2$ of the $\cos 2\phi'$ efficiency harmonic between the $m = \pm 1$ -rich and $m = 0$ -rich samples enters $\hat{A}$ as $\delta\epsilon_2/(P_+ - P_0)$, verified against the pseudo-experiments (`reco.fill_acceptance_bias`): $\delta\epsilon_2 = 10^{-3}$ fakes 5.6×10^{-4} , which is 5% of the sweet-spot amplitudes but 4.6 times the one-year statistical error and 14 times the ten-year one. Bunch-by-bunch alternation, or stability of the harmonic to $\approx 10^{-4}$ between fills, is therefore a requirement of the measurement. The coherent counterpart is *tighter* than the earlier estimate: a naive $\delta(\cos 2\beta)/(P_+ - P_0)$ gives 1.3% of a_t for a 1% change of the vertical envelope between fills, but the fit gives +19% for a 10^{-3} change and +169% for 1% — under the slot only half the β bins are live and the t-shape template is 99% anti-correlated with the constant there, so a shape perturbation is amplified about a hundredfold. The requirement is bunch-by-bunch alternation or $\approx 10^{-4}$ envelope stability; the mitigation, if neither is available, is that the higher- $|t|$ bins are far less exposed (+3.9% and +0.04% at the same 10^{-3} for $|t| = 0.08\text{--}0.12$ and $0.12\text{--}0.17$ GeV², where more of the circle is live). K is model-dependent at the 3–11% level between the repository's Δ shapes (moment_A/moment_B/toy at the four spots), larger than the 10-year statistical errors; the EMCAL term $2\%/\sqrt{E} \oplus 1\%$ is the backward specification applied at all η (the sweet-spot electrons are backward, the high- x bins of the amplitude panels are not); the 6R sample is background-free and Z-identified by assumption. u_1 , u_2 and the luminosity shares are exactly known in Table 3, but no longer only by assumption: they are arguments of the fit (`measure_coherent(u_coeffs_assumed=, lumi_assumed=, responses=)` , `money_cos2phi_coherent_reco.py --u2-assumed --rel-lumi-offset --envelope-split`), and the sensitivities above are what they produce. The relative luminosity is second order here as it is inclusively (10^{-3} moves a_t by 0.08%).

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- [18] ATHENA Collaboration, J. Adam et al., *ATHENA detector proposal — a totally hermetic electron nucleus apparatus proposed for IP6 at the Electron-Ion Collider*, JINST 17 (2022) P10019, arXiv:2210.09048, Sec. 3.1 and Fig. 22 (best reconstruction method and y resolution over the (x, Q^2) plane at 18×275 GeV; e^-Z/DA for $y \lesssim 0.1$; JB 20–30%).
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simulation at $Q^2 > 100 \text{ GeV}^2$).

[20] M. Arratia, D. Britzger, O. Long, B. Nachman, *Reconstructing the kinematics of deep inelastic scattering with deep learning*, Nucl. Instrum. Meth. A 1025 (2022) 166164, arXiv:2110.05505 (Table 1: all classical methods; Fig. 5: RMS of Q^2 , y , x versus y for the ATHENA fast simulation; Sec. 5: acceptance and noise effects on the hadronic methods at low y). Related: R. Aggarwal, A. Caldwell, *Kinematic fitting of neutral current events in deep inelastic ep collisions*, JINST 17 (2022) P09035, arXiv:2206.04897. Local copies of [9], [12]–[20] are indexed in `refs/README.md` and `refs/refs_dict.json`.

Appendix A • Revision history

DATE	REVISION
2026-08-24	First version: §§1–6 — what is measured, the covariant azimuth and the head-on frame, the inclusive and coherent chains, the audit of the simulation and the recommendations (Figures 1–2, Schematics 1–3, <code>polligen.reco</code>). Same day: §7 — the reconstructed-level pseudo-experiments of WP3 (money plots 5R, 7R, 6R from <code>polligen.recopseudo</code> ; the acceptance-weighted template basis; the anchor's $2a_2$ convention verified from the paper's text).
2026-08-25	Revised against the reference copies in refs/: the ePIC Roman-Pot slot geometry and beam-effect resolution [17], the ZEUS bounds on the diffractive harmonics [13,14], the exact spin-1 harmonics at finite γ [12], the anchor's frame and convention [9]; the hadronic- y resolution set to the documented 25% [18–20] and money plots 5R/7R rerun. Corrections and the further known limits of §7.1 from the code review (docs/code_review_2026-08-25.md).
2026-08-25	Hadronic final state added to the chain (WP3-HFS): <code>polligen.hfs</code> — the Σ /JB/DA methods on a generated final state, the hadron-side detector response, a toy fragmentation stand-in and the (x, Q^2) event library; the PYTHIA 8 generation script for eic-shell; §3 and Figure 3; the §7 cross-check of Table 2 with the HFS-based y .
2026-08-25	Editorial pass: the note restated as current content. The audit table and §6 give today's status rather than the history of the code; the implementation recipe of the old §5.1 is dropped in favour of §7; the duplication between the lede, the verdict and the recommendations is removed.

Conventions: head-on frame with the ion along $+z$ and the electron along $-z$; θ' is the scattered-electron angle from the electron beam; per-nucleon (x, y, s) throughout; whole-nucleus four-momenta where a mass matters. Beam divergences are the proton values implied by the documented Roman-Pot cuts (73 and $149 \mu\text{rad}$); the lithium optics are undocumented (plans/04 #11). Built by `reports/build_report.py`; figures from the `evgen/scripts` named in the text; display mathematics typeset at build time.